

TROY LORENTS

Full-Stack Developer & AI Engineer

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PROFESSIONAL SUMMARY

Full-stack and AI engineer with 8+ years of building production, cloud-native systems and the last 5+ years focused on AI-powered software, intelligent automation, and enterprise workflow optimization. Experienced delivering end-to-end solutions across React/TypeScript frontends, Python (FastAPI) and .NET backends, and scalable data platforms. Designed and deployed multi-agent systems, RAG pipelines, and production AI platforms using Azure AI Foundry, OpenAI, and Claude to automate business processes, reduce operational bottlenecks, and improve organizational efficiency.

TECHNICAL SKILLS

AI & ML: Azure OpenAI, Azure AI Foundry, Azure AI Search, Azure Document Intelligence, Model Context Protocol (MCP) Integrations, Agentic & Multi-Agent Systems, RAG & Agentic RAG, Hybrid Search (BM25 + Vector), LangChain/LangGraph, Semantic Kernel, PyTorch, Large Language Models

AI Development & Evaluation: Codex, Claude Code, Cursor, GitHub Copilot, Autonomous Coding Agents, Model Context Protocol (MCP), Agent Context Engineering, AGENTS.md, Custom Agent Skills, LLM and Agent Evaluation Harnesses, Terminal-Bench 2.1, AI-Assisted Testing and Debugging

Languages: Python, C#, TypeScript, JavaScript, SQL, Bash, PowerShell

Frontend: React, Next.js, Vue, React Native, TanStack, Material UI, Tailwind CSS

Backend: Python, FastAPI, Flask, .NET Core, Node.js/Express, REST, GraphQL, SDK Design & Development, RBAC (Role-Based Access Control), OAuth 2.0 and SSO integration.

Data: SQL Server, PostgreSQL, MySQL, Cosmos DB, DynamoDB, MongoDB, Neo4j, Redis, Qdrant, REST APIs

Cloud & DevOps: AWS (EC2, EKS, ECS/Fargate, Lambda, API Gateway, S3, CloudWatch, Amplify, Elastic Beanstalk), Azure (App Service, Azure DevOps, API Management, AI Search, Cosmos DB), Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI, CircleCI, Octopus Deploy, Rancher, Cloudflare

Automation & Integration: n8n, Power Automate, Jira, Confluence, UiPath, ServiceNow, Airtable, Stripe

PROFESSIONAL EXPERIENCE

Arizona State University | Web Application Developer (Full-Stack & AI) 2024 - Present

School of Computing & Augmented Intelligence, Ira A. Fulton Schools of Engineering

- Built SAMS, an end-to-end student hiring platform replacing a 150–200-hour manual process; serves 300+ users, cut processing time ~80%, and has expanded from SCAI to multiple schools across Ira A. Fulton Schools of Engineering. Led rollout backed by HR leadership and the Dean's Office.
- Re-architected AWS hosting onto a single EC2 Windows Server instance running 600+ student .NET API projects, each team container-isolated, each professor having their own admin view to test and grade the student's projects, cutting the monthly bill from \$850 to \$60 (93% reduction).
- Developed and integrated a production conversational hiring agent into SAMS using Python, FastAPI, and LLM APIs. The agent answers hiring and policy questions, retrieves and delivers relevant documents, and uses provider-agnostic RAG, vector search, tool-calling workflows and model routing across 17+ LLMs to support live users.
- Designed and maintained SQL Server, PostgreSQL, and Airtable DB data models supporting transactional and AI-driven workloads.

- Deployed and operated services across AWS EC2, Elastic Beanstalk, EKS, ECS/Fargate, and Lambda using Elastic Load Balancing, Auto Scaling, CloudWatch monitoring, Pytest, load testing, and deployment-validation loops; also deployed services through Azure DevOps, App Service, and API Management.
- Implemented modern authentication and authorization using OAuth 2.0, CAS SSO, and RBAC across enterprise applications and AI services, enabling secure API access and role-based permissions.

Zywave | Software Engineer 2018 - 2024

Healthcare Data SaaS, Scottsdale, Arizona

- Engineered enterprise healthcare SaaS with React, TypeScript, C#, ASP.NET Core, Entity Framework, and Dapper, supporting 10,000+ users with HIPAA-compliant architecture and secure API standards.
- Built an API quoting adapter for an insurance carrier in C#/ASP.NET Core using LINQ, Dapper, Swagger/OpenAPI, with comprehensive unit-test coverage using Moq and Fluent Assertions.
- Designed databases across MySQL, SQL Server, MariaDB, and DynamoDB; tuned SQL and async processing to reduce latency and production incidents.
- Owned AWS infrastructure across EC2, S3, Lambda, DynamoDB, CloudWatch, supporting containerized deployments and CI/CD across QA, PRE, and DEV environments using Docker, Rancher, GitLab, CircleCI, and Octopus Deploy.

RateFactory (acquired by Zywave) | Data Administrator 2017 - 2018

- Built and ran ETL pipelines on large insurance datasets with Python (Pandas, OpenPyXL), Excel VBA, and uBot Studio web scrapers.
- Led a four-person team, owning task planning, onboarding, and QA across SQL Server, PostgreSQL, and DynamoDB.

PROJECTS

- **Coding Agent Evaluation Harness:** Built a 735-line autonomous coding agent and evaluation harness, then independently ran the complete 89-task Terminal-Bench 2.1 benchmark using a frontier model, achieving a 61.8% score. Implemented task execution, environment management, tool orchestration, result collection, and automated evaluation.
- **useKnockout API:** Built a production serverless GPU inference API using FastAPI, Modal, PyTorch, BiRefNet, and ProRes 4444 delivering automatic scale-to-zero deployments and published multi-language SDKs for Node.js, Python, React, and CLI to npm and PyPI. www.useknockout.com
- **Agent OS:** Open-source multi-agent orchestration platform built with FastAPI and React/TypeScript, supporting autonomous multi-agent workflows, secure REST APIs, PyTorch-based ML experimentation, Cloudflare, Neo4j, Redis, and Qdrant for graph, memory, and vector retrieval.
- **Resume Match AI:** Designed and built a full-stack AI platform using React, Flask, Azure OpenAI, Azure AI Search, Azure Cosmos DB, and Azure Document Intelligence, implementing production RAG pipelines, hybrid BM25 + vector search with RRF ranking, semantic skill extraction, embeddings, and REST APIs for intelligent resume-to-job matching.
- **ATLAS:** Architected a multi-tenant AI SaaS platform using Python (FastAPI), React, PostgreSQL, Azure OpenAI, and specialized AI agents to automate business operations through role-based workflows, human-in-the-loop approvals, and cloud-native microservices.

Additional projects at troy-lorents-portfolio.netlify.app

EDUCATION

B.S. Software Engineering, Arizona State University (Expected May 2027)

B.A. Business Law, Arizona State University (May 2018)